

| Question number | Answer                                                                                                                                                                                                                                                                                                                   | Additional guidance                                                                                                                                                                                                                                                                                                                            | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 8(a)            | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>because the {phenotype / blood group} of individual 3 is {different / combination} from that of individual 1 and 2 (1)</li> <li>because neither of the alleles are dominant / both alleles are expressed (1)</li> </ul> | <p><b>ACCEPT</b> the phenotype of the heterozygote is different from the phenotype of either homozygote<br/> MM gives you blood group M + NN gives you blood group N but MN gives you a different blood group<br/> offspring have both the blood types (in their phenotype)</p> <p><b>ACCEPT</b> both alleles are equal / equally dominant</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                 | Additional guidance                                                                                                                                                                                                                                                                                        | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 8(b)            | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• genotypes of individuals 3 and 4 shown / alleles in gametes of individuals 3 and 4 shown (1)</li> <li>• possible genotypes of children shown (1)</li> <li>• corresponding phenotypes given (1)</li> <li>• 1 type M : 1 type N : 2 type MN (1)</li> </ul> | <p>genotypes : MN / MN      gametes : M or N / M or N</p> <p>MM and NN and 2 × MN</p> <p>MM = blood group M and NN = blood group N and MN = blood group MN<br/> <b>DO NOT ACCEPT</b> blood group MM / NN</p> <p><b>ACCEPT</b> in any order provided it is clear which phenotype each number relates to</p> | (4)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Additional guidance                                                                                             | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------|
| 8(c)(i)         | <p>An explanation that includes three of the following points:</p> <ul style="list-style-type: none"> <li>individual 3 will have both (M and N) proteins (on the surface the of the red blood cells) (1)</li> <li>because individual 3 will {be heterozygous / have one allele for M and one allele for N / have the genotype MN} (1)</li> <li>(both) genes will be transcribed / mRNA produced for (both) genes (1)</li> <li>and translated (1)</li> </ul> | <p><b>ACCEPT</b> one gene for M and one gene for N</p> <p><b>ACCEPT</b> {alleles / DNA} will be transcribed</p> | <b>(3)</b> |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| *8(c)(ii)       | <p><b>Substitution mutation:</b></p> <ul style="list-style-type: none"> <li>• substitution swaps one base for another</li> <li>• may not alter the amino acid coded for</li> <li>• as the genetic code is degenerate</li> <li>• triplet codon codes for the same amino acid</li> <li>• therefore, no effect on the protein</li> <li>• therefore, no effect on the phenotype</li> <li>• amino acid could be different</li> <li>• therefore, shape of protein may or may not be (significantly) different</li> <li>• depending on significance the phenotype may stay the same</li> <li>• depending on the significance the phenotype may become the same as one of the parents</li> <li>• a stop codon could be coded for</li> <li>• so the protein may be {shorter / not coded for}</li> <li>• so the phenotype may become the same as one of the parents</li> </ul> <p><b>Insertion / deletion mutation:</b></p> <ul style="list-style-type: none"> <li>• deletion removes one base</li> <li>• insertion adds in a base</li> <li>• causing a frameshift</li> <li>• this will probably have a huge effect on protein</li> <li>• phenotype may become one of parents</li> <li>• may have little effect on protein if near the end of the gene sequence</li> <li>• and therefore, have very little effect on phenotype</li> </ul> | (6)  |

|         |     |                                                                                                                                                                                                                                                                                                                                                                                                           | Additional guidance                                                                                                                                                                                                                                                                                                                               |
|---------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 0 | 0   | No awardable content                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                   |
| Level 1 | 1-2 | Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context. | Simple links made between mutation and effect on protein or phenotype<br><br>1 mark = one comment<br><br>2 marks = three comments                                                                                                                                                                                                                 |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.                                        | Extended links made between mutation and effect on protein or effect on phenotype in the context of the question<br><br>3 marks = extended comments about one effect of one type of mutation<br><br>4 marks = extended comments about one effect of two types of mutations<br>OR<br>extended comments about two effects of one mutation           |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.         | Clear discussions on all aspects of mutations and effect on protein and effect on phenotype in the context of the question<br><br>5 marks = extended comments about one effect of one type of mutation<br><br>6 marks = extended comments about one effect of two types of mutations<br>OR<br>extended comments about two effects of one mutation |